Prime-Complete Neighbours of Perfect Powers

A conjecture, computational evidence, and a residue-class obstruction

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Abstract

A positive integer is *prime-complete* if its prime divisors are exactly the first r primes for some r. We conjecture that for $k \geq 3$ and $y \geq 2$ the numbers $y^k - 1$ and $y^k + 1$ are prime-complete only for $7^4 - 1 = 2400 = 2^5 \cdot 3 \cdot 5^2$. For even k the conjecture reduces exactly to a subcase of the OEIS sequence A141399 (prime-complete products $m(m+1)$), and holds for $y^{k/2} < 9.49 \times 10^{29}$. For odd k we prove a residue-class obstruction: the primes $q \leq p_r$ with $\gcd(q-1, k) = 1$ must all divide $y \mp 1$, which forces $y \geq P_r^{\delta_k + o(1)}$ and forces the cyclotomic cofactor to carry average prime exponent at least $(k-1)\delta_k/(1-\delta_k)$ (equal to 2 for $k = 3$). Using Bennett's theorem on $|ax^n - by^n| = 1$ we show that at most $k^r - 1$ values of y make $y^k \pm 1$ smooth over the first r primes, and we classify them completely for $k = 3$, $r \leq 8$ (unconditionally for $r \leq 6$, under GRH for $r = 7, 8$). Assuming the *abc* conjecture, the conjecture holds up to finitely many exceptions for every fixed odd $k \geq 5$; the case $k = 3$ is exactly *abc*-critical, and is settled under *abc* together with an explicit lower bound for the least nontrivial cube root of ± 1 modulo the primorial, which holds with room on all computed data. All prime-complete numbers below 10^{30} (about 4.6×10^8 of them) were tested; the only example is 2400.

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*The conjecture was proposed by Claude in the course of a collaboration with Ken Clements on OEIS A141399 (October 2026). Clements supplied the A141399 framework, the certified A141399 range quoted in Corollary 2.3, and the OEIS generator that the enumeration program modifies. All computations reported here were run by Claude. Attribution and authorship of any circulated version are to be settled by K. Clements.

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1 Introduction

Let p_r be the r-th prime, $P_r = p_1 p_2 \cdots p_r$ the r-th primorial, $\theta(x) = \sum_{p \le x} \log p$ and $\mathrm{rad}(n) = \prod_{p \mid n} p$.

Definition 1.1. An integer $N \ge 2$ is *prime-complete of order* r if $\mathrm{rad}(N) = P_r$.

The prime-complete numbers form OEIS A055932 [1]. The sequence A141399 [2] lists the m for which $m(m+1)$ is prime-complete; it has 28 known terms, the largest $m = 633555$, and its termination is an open problem that contains the classical question of which primorials are products of two consecutive integers [13]. Equivalently, A141399 asks when $x^2 - 1$ is prime-complete for odd $x = 2m + 1$.

Clements has conjectured that A141399 is the "poster child" of a general phenomenon: a polynomial generator with integer roots that is not a constant multiple of a perfect power should have only finitely many prime-complete values. Polynomials that split into linear factors give S-unit equations, which is why A141399 can be attacked by Pell equations. This note studies the simplest family that is *not* split but is still *intersective* (has a root modulo every prime): the polynomials $x^k \pm 1$.

Conjecture 1.2. *Let $y \ge 2$ and $k \ge 3$. Then $y^k - 1$ and $y^k + 1$ are prime-complete only for*

$$7^4 - 1 = 2400 = 2^5 \cdot 3 \cdot 5^2 \qquad (r = 3).$$

Results.

1. *Even k* (§2). $y^k + 1$ is never prime-complete, and $y^k - 1$ is prime-complete exactly when $m = (y^{k/2} - 1)/2$ is a term of A141399. The conjecture therefore holds for even k whenever $y^{k/2} < 9.49 \times 10^{29}$.

2. *Odd k: a residue-class obstruction* (§3). Lemma 3.1 forces $y \equiv \pm 1$ modulo every prime $q \le p_r$ with $\gcd(q - 1, k) = 1$, and Proposition 3.3 shows that the remaining cyclotomic factor must then be "nearly powerful" on the other residue classes.

3. *Counting and exact classification* (§4). By Bennett's theorem, at most $k^r - 1$ integers $y \ge 2$ make $y^k \pm 1$ smooth over the first r primes. For $k = 3$ we list them all for $r \le 8$, and a squeeze argument modelled on A141399 shows the cubic "wall" is closed at every computed order.

4. *Conditional results* (§5). The *abc* conjecture implies finiteness for each fixed odd $k \ge 5$. The case $k = 3$ is exactly *abc*-critical; *abc* plus a lower bound for the cubic floor settles it.

5. *Evidence* (§7) and *a programme for a proof* (§8).

2 Even exponents

Proposition 2.1. *If $k \ge 2$ is even and $y \ge 2$, then $y^k + 1$ is not prime-complete.*

Proof. Since y^k is a square, $y^k \equiv 0$ or $1 \pmod 3$, so $3 \nmid y^k + 1$. A prime-complete number not divisible by 3 has order 1, i.e. $y^k + 1 = 2^a$. Then y is odd, $y^k \equiv 1 \pmod 8$, $y^k + 1 \equiv 2 \pmod 8$, so $y^k + 1 = 2$, impossible for $y \geq 2$. □

Proposition 2.2. *Let $k = 2j$ with $j \geq 2$ and $y \geq 2$. Then $y^k - 1$ is prime-complete if and only if y is odd and $m := (y^j - 1)/2$ is a term of A141399; the orders agree.*

Proof. If y is even, $y^k - 1$ is odd and greater than 1, hence not prime-complete. If y is odd, put $x = y^j = 2m + 1$. Then $y^k - 1 = x^2 - 1 = 4m(m + 1)$, and since $m(m + 1)$ is even, $\mathrm{rad}(4m(m + 1)) = \mathrm{rad}(m(m + 1))$. □

Corollary 2.3. *Among the 28 known terms of A141399 the only m with $2m+1$ a perfect power is $m = 24$ ($2m + 1 = 49 = 7^2$), which gives $7^4 - 1 = 2400$. A capped Pell enumeration of all consecutive p_{35}-smooth pairs with $m \leq U_{35}$ (Clements, 2026, unpublished; a prime-complete pair with $m(m + 1) < P_{36}$ has order at most 35) shows that A141399 has no term m with $633555 < m \leq U_{35} = 4746783489815299942493786409 41$. Hence Conjecture 1.2 holds for every even k with $y^{k/2} \leq 2U_{35} + 1 \approx 9.49 \times 10^{29}$, and for all even k if A141399 has no further terms.*

So the even case is a refinement of A141399 (a term whose Pell abscissa $2m + 1$ is a perfect power) and contains nothing new. The new content of the conjecture is odd k.

3 Odd exponents: the residue-class obstruction

Throughout this section $k \geq 3$ is odd. Define

$$Q_k(r) = \prod_{\substack{q \leq p_r \\ \gcd(q-1,k)=1}} q, \qquad \delta_k = \prod_{\ell \mid k} \frac{\ell - 2}{\ell - 1} \quad (\ell \text{ prime}).$$

Note that $2 \mid Q_k(r)$, and $3 \mid Q_k(r)$ once $p_r \geq 3$, for every odd k (since $q - 1 \in \{1, 2\}$).

Lemma 3.1 (complementary classes divide $y \mp 1$)**.** *Suppose $y^k - 1$ (resp. $y^k + 1$) is prime-complete of order r. Then y is odd, $\gcd(y, P_r) = 1$, and*

$$Q_k(r) \mid y - 1 \qquad (\textit{resp. } Q_k(r) \mid y + 1).$$

In particular $y \geq Q_k(r) + 1$ (resp. $y \geq Q_k(r) - 1$).

Proof. Every $q \leq p_r$ divides $y^k \mp 1$, so $q \nmid y$; and $2 \mid y^k \mp 1$ forces y odd, so $2 \mid y \mp 1$. Let $q \leq p_r$ be odd with $\gcd(q - 1, k) = 1$, and let $t = \mathrm{ord}_q(y)$, so $t \mid q - 1$. For $y^k \equiv 1$: $t \mid \gcd(k, q - 1) = 1$, so $y \equiv 1 \pmod q$. For $y^k \equiv -1$: $t \mid 2k$ and $t \nmid k$; since $\gcd(2k, q - 1) \mid 2$, we get $t = 2$ and $y \equiv -1 \pmod q$. □

Lemma 3.2. *For fixed odd k, $\log Q_k(r) = (\delta_k + o(1))\, \theta(p_r)$ as $r \to \infty$.*

Proof. For $q \nmid k$, the condition $\gcd(q - 1, k) = 1$ says $q \not\equiv 1 \pmod \ell$ for each prime $\ell \mid k$; it depends only on q modulo $\mathrm{rad}(k)$ and holds in a proportion $\prod_{\ell \mid k} (\ell - 2)/(\ell - 1)$ of the reduced classes. Apply the prime number theorem in arithmetic progressions. □

Proposition 3.3 (valuation demand)**.** *Suppose $y^k - 1$ is prime-complete of order r and put $C(y) = (y^k - 1)/(y - 1)$. Every prime factor q of $C(y)$ satisfies $q \le p_r$ and either $q \mid k$ or $\gcd(q-1, k) > 1$, and $C(y) > Q_k(r)^{k-1}$. The same holds for $y^k + 1$ with $C(y) = (y^k + 1)/(y+1)$ and the bound $C(y) \ge (Q_k(r) - 2)^{k-1}$.*

Proof. Let $q \le p_r$ with $q \nmid k$ and $\gcd(q-1, k) = 1$. By Lemma 3.1, $y \equiv 1 \pmod q$, so $C(y) = \sum_{i<k} y^i \equiv k \not\equiv 0 \pmod q$. The size bound follows from $C(y) > y^{k-1}$ and $y > Q_k(r)$. The y^k+1 case is identical with $y \equiv -1$, $C(y) = \sum_{i<k}(-y)^i$ and $C(y) \ge y^{k-2}(y-1) \ge (y-1)^{k-1}$. □

Corollary 3.4 (the case $k = 3$)**.** *If $y^3 - 1$ is prime-complete of order r, then $\Phi_3(y) = y^2 + y + 1$ equals $3 \prod_{q \equiv 1\,(3)} q^{v_q}$ with all $q \le p_r$, and*

$$\sum_{\substack{q \le p_r \\ q \equiv 1\,(3)}} v_q \log q \;>\; 2 \log Q_3(r) - \log 3.$$

Consequently the $\log q$-weighted mean of the exponents v_q over all primes $q \equiv 1 \pmod 3$ up to p_r is at least

$$\mu_3(r) := \frac{2 \log Q_3(r) - \log 3}{\theta(p_r; 3, 1)} \;\longrightarrow\; 2.$$

The same holds for $y^3 + 1$ with $\Phi_6(y) = y^2 - y + 1$ and $Q_3(r) - 2$ in place of $Q_3(r)$.

Proof. Here $y \equiv 1 \pmod 3$, so $\Phi_3(y) \equiv 3 \pmod 9$ and $v_3(\Phi_3(y)) = 1$. The limit follows from Lemma 3.2 with $\delta_3 = \frac{1}{2}$ and $\theta(x; 3, 1) \sim \frac{1}{2}\theta(x)$. □

For general odd k the same argument gives a mean exponent of at least $(k-1)\delta_k/(1-\delta_k) + o(1)$ on the admissible classes: 2 for $k = 3$, 12 for $k = 5$, 30 for $k = 7$, 8 for $k = 9$. Table 1 shows the convergence.

Table 1: The residue-class tax. $\log Q_k/\theta$ is the proved lower bound for $\log y / \log P_r$ (Lemma 3.1); μ_3 is the required mean exponent of $\Phi_3(y)$ (Corollary 3.4). Limits: $\delta_3 = \frac{1}{2}$, $\delta_5 = \frac{3}{4}$, $\delta_7 = \frac{5}{6}$, $\mu_3 \to 2$.

p_r	r	$\log Q_3/\theta$	μ_3	$\log Q_5/\theta$	$\log Q_7/\theta$
47	15	0.554	2.43	0.767	0.826
97	25	0.525	2.18	0.786	0.864
199	46	0.524	2.19	0.773	0.861
997	168	0.518	2.15	0.758	0.827
9973	1229	0.501	2.01	0.750	0.834
99991	9592	0.501	2.01	0.751	0.834

Remark 3.5 (why $k = 2$ pays no tax). For $x^2 - 1 = (x-1)(x+1)$ both factors are linear, so every prime may divide either, and no residue class is forced into a factor that is too small to hold it. This is the precise sense in which A141399 is the *hardest* member of the family $x^k \pm 1$. Each non-linear irreducible factor of an intersective polynomial is supported on Frobenius-restricted classes and pays a tax of this kind.

Example 3.6. "Nearly square" values of Φ_3 are plentiful: the Pell family $(2y+1)^2 - 12z^2 = -3$ gives $y^2 + y + 1 = 3z^2$ infinitely often, e.g. $y = 22$, $\Phi_3(22) = 507 = 3 \cdot 13^2$. So the obstruction is not powerfulness by itself. It is powerfulness *on one residue class*, combined with the size forced by Lemma 3.1.

4 Counting and exact classification

Theorem 4.1 (Bennett [3])**.** *If a, b, n are positive integers with $n \geq 3$, the equation $|ax^n - by^n| = 1$ has at most one solution in positive integers.*

Corollary 4.2. *For $k \geq 3$ and $r \geq 1$, at most $k^r - 1$ integers $y \geq 2$ have $y^k - 1$ or $y^k + 1$ smooth over $\{p_1, \ldots, p_r\}$.*

Proof. Write $y^k \mp 1 = dw^k$ with d k-th-power-free and p_r-smooth; there are k^r such d. Then (y, w) is a positive solution of $|x^k - dY^k| = 1$, so each d yields at most one y, across both signs. For $d = 1$ there is none. □

For $k = 2$ the corresponding count is weaker: Lehmer's method allows up to $\max(3, (p_r+1)/2)$ solutions per Pell class [10]. For $k \geq 3$ each class carries at most one, and it is found by solving one Thue equation. For $k = 3$ this is the classical theorem of Delone and Nagell (see [12] [verify: chapter]), and since $x - Y\sqrt[3]{d}$ is a unit of a field of unit rank one, the solution is a power of its fundamental unit.

Algorithm. For each cube-free p_r-smooth $d > 1$, solve $x^3 - dY^3 = 1$ (PARI/GP `thue` [15], which implements the Bilu–Hanrot method [4]). Each solution with $Y \neq 0$ and $|x| \geq 2$ gives $y = |x|$ with $y^3 - 1 = dY^3$ ($x > 0$) or $y^3 + 1 = d|Y|^3$ ($x < 0$). Keep those with Y p_r-smooth.

Theorem 4.3 (computation)**.** *For $r \leq 6$ unconditionally, and for $r \leq 8$ assuming the generalized Riemann hypothesis (used by the class-group computations inside `thue`), the integers $y \geq 2$ with $y^3 - 1$ or $y^3 + 1$ smooth over the first r primes are exactly those in Table 2. None is prime-complete.*

Independently, a direct search over $y \leq 3 \times 10^6$ finds exactly the same 21 values for 19-smooth $y^3 \pm 1$.

Table 2: All $y \geq 2$ with $y^3 \mp 1$ smooth over the primes up to 19 ($r \leq 8$). "Order" is $\pi(P^+(y^3 \mp 1))$.

y	form	d	order	y	form	d	order
2	y^3+1	9	2	11	y^3-1	1330	8
2	y^3-1	7	4	12	y^3+1	1729	8
3	y^3+1	28	4	16	y^3-1	4095	6
3	y^3-1	26	6	17	y^3+1	182	6
4	y^3+1	65	6	18	y^3-1	17	7
4	y^3-1	63	4	19	y^3+1	20	4
5	y^3+1	126	4	22	y^3-1	10647	6
7	y^3-1	342	8	23	y^3+1	1521	6
8	y^3+1	19	8	31	y^3+1	3724	8
9	y^3-1	91	6	69	y^3+1	328510	8
10	y^3+1	1001	6				

4.1 The cubic squeeze

The method that controls A141399 at small orders transfers directly. Define the *cubic floor* and *cubic ceiling*

$$A_3^{\mp}(r) = \min\{y > 1 : \ y^3 \equiv \pm 1 \pmod{P_r}\},$$
$$L_3(r) = \max\{y \geq 2 : \ y^3 - 1 \text{ or } y^3 + 1 \text{ is } p_r\text{-smooth}\}.$$

A prime-complete $y^3 \mp 1$ of order r satisfies $A_3^{\mp}(r) \leq y \leq L_3(r)$, the analogue of A118478$(r) \leq m \leq$ A002072(r) for A141399.

The floor is cheap to compute exactly. By Lemma 3.1, $y \equiv \pm 1 \pmod{Q_3(r)}$, and modulo each prime $q \equiv 1 \pmod 3$ there are three admissible classes, so there are $3^{\pi(p_r;3,1)}$ roots to compare. The ceiling is the expensive side (Theorem 4.3).

Table 3: The cubic squeeze. The wall $A_3 > L_3$ holds at every order, unlike A141399, where it first closes at $r = 14$.

r	p_r	roots	$A_3^-(r)$	$A_3^+(r)$	$L_3(r)$
1	2	1	3	3	–
2	3	1	7	5	2
3	5	1	31	29	2
4	7	3	121	59	19
5	11	3	331	1319	19
6	13	9	991	4289	23
7	17	9	145861	50489	23
8	19	27	241231	639539	69

The floor was computed exactly through $r = 32$. Writing $\alpha^{\mp}(r) = \log A_3^{\mp}(r)/\log P_r$, we find

$$\min_{4 \leq r \leq 32} \alpha^{\mp}(r) = 0.669 \ (r = 6), \qquad \min_{14 \leq r \leq 32} \alpha^{\mp}(r) = 0.806,$$

against the proved bound $\log Q_3(r)/\log P_r \approx \frac{1}{2}$ of Lemma 3.1. Heuristically the least of $3^{\pi(p_r;3,1)} = P_r^{o(1)}$ roots lies near $P_r^{1-o(1)}$, so $\alpha(r) \to 1$.

5 Conditional results

Recall the *abc* conjecture [11, 14]: for every $\varepsilon > 0$ there is K_ε such that coprime positive $a+b = c$ satisfy $c < K_\varepsilon \operatorname{rad}(abc)^{1+\varepsilon}$.

Lemma 5.1. *For odd $k \geq 5$, $(k-1)\delta_k > 1$; for $k = 3$, $(k-1)\delta_k = 1$.*

Proof. Each factor $(\ell-2)/(\ell-1) \geq \frac{1}{2}$ and $3^{\omega(k)} \leq k$, so $\delta_k \geq 2^{-\omega(k)} \geq k^{-\log 2/\log 3}$. The function $(k-1)k^{-\log 2/\log 3}$ is increasing and equals 1.45 at $k = 5$. For $k = 3$, $\delta_3 = \frac{1}{2}$. □

Theorem 5.2. *Assume abc. For each fixed odd $k \geq 5$, only finitely many pairs (y, r) have $y^k - 1$ or $y^k + 1$ prime-complete of order r.*

Proof. Apply *abc* to $(1, y^k - 1, y^k)$ or $(1, y^k, y^k + 1)$. The radical is $P_r \operatorname{rad}(y) \le P_r y$, so

$$(k - 1 - \varepsilon) \log y < (1 + \varepsilon)\, \theta(p_r) + \log K_\varepsilon. \tag{1}$$

By Lemmas 3.1 and 3.2, $\log y \ge (\delta_k - \eta)\theta(p_r)$ for $r \ge r_1(k, \eta)$. Substituting, $\big[(k - 1 - \varepsilon)(\delta_k - \eta) - (1 + \varepsilon)\big]\theta(p_r) < \log K_\varepsilon$. By Lemma 5.1 the bracket is positive for small ε, η, so r is bounded; for each r, (1) bounds y. □

Proposition 5.3 ($k = 3$ is *abc*-critical)**.** *Under abc,* (1) *gives* $\log y \le (\frac{1}{2} + O(\varepsilon))\theta(p_r) + O_\varepsilon(1)$*, while Lemma 3.1 gives* $\log y \ge (\frac{1}{2} - o(1))\theta(p_r)$*. The window is nonempty for every* $\varepsilon > 0$*, so abc alone does not decide* $k = 3$*. The same happens with Langevin's consequence of abc [9, 6],* $\operatorname{rad}(y^3 \mp 1) \gg_\varepsilon y^{2-\varepsilon}$*.*

Theorem 5.4. *Assume abc, and assume there is* $c > 0$ *with*

$$\log A_3^{\mp}(r) \;\ge\; \big(\tfrac{1}{2} + c\big)\, \theta(p_r) \qquad \text{for all large } r. \tag{F$_3$}$$

Then only finitely many (y, r) *have* $y^3 \mp 1$ *prime-complete of order* r*.*

Proof. A prime-complete $y^3 \mp 1$ of order r has $y \ge A_3^{\mp}(r)$, so (F$_3$) and (1) with $k = 3$ give $(2 - \varepsilon)(\frac{1}{2} + c)\theta < (1 + \varepsilon)\theta + O_\varepsilon(1)$, which fails for large r once $\varepsilon < c/(1 + c)$. □

Remark 5.5 (comparison with A141399). For A141399, *abc* caps m at $P_r^{1+\varepsilon}$, while even the expected size $P_r^{1-o(1)}$ of the floor A118478(r) lies below that cap; so *abc* plus any floor bound fails to close it, and a support-sensitive strengthening of *abc* is needed. For $k = 3$, *abc* caps y at $P_r^{1/2+\varepsilon}$ while the computed floor exponent is above 0.8, so *abc* plus a modest floor bound suffices. In this precise sense the cubic case is strictly easier than A141399.

Remark 5.6 (*xyz* applies only partially). By Lemma 3.1, every prime factor of y exceeds p_r, so the triple $(1, y^k \mp 1, y^k)$ is never p_r-smooth. The Lagarias–Soundararajan *xyz* conjecture [8], which implies finiteness of A141399, controls only those solutions in which y has no large prime factor, and nothing in the problem prevents $P^+(y)$ from being large.

Table 4: What each conjecture decides.

Case	*abc*	*xyz*
$k = 2$ (A141399)	silent (quality < 1)	finite
even $k \ge 4$	as A141399	finite (via A141399)
$k = 3$	critical; finite with (F$_3$)	only when y is smooth
odd $k \ge 5$, fixed	finite (Theorem 5.2)	only when y is smooth

6 A heuristic expected count

Model y as random among integers with $y^k \equiv 1 \pmod{P_r}$, a set of density $G_k(r)/P_r$ with $G_k(r) = \prod_{q \le p_r} \gcd(k, q - 1) = P_r^{o(1)}$, and model the cofactor $t = (y^k - 1)/P_r$ as a random integer of its size. In the relevant range $\log t \asymp p_r$, so the probability that t is p_r-smooth is $t^{-1+o(1)}$ (de Bruijn [5]; see also [7]). Summing over dyadic ranges $Y \ge Q_k(r)$,

$$E_k(r) \approx \sum_Y \frac{Y\, G_k(r)}{P_r} \Big(\frac{Y^k}{P_r}\Big)^{-1+o(1)} = \exp\Big(-\big((k - 1)\delta_k + o(1)\big)\theta(p_r)\Big),$$

dominated by $Y \approx Q_k(r)$. For $k = 3$ this is $P_r^{-1+o(1)}$, compared with $P_r^{-1/2+o(1)}$ for the same model applied to A141399. Given the exact classification through $r = 8$, the expected number of cubic examples at all larger orders is about $\sum_{r \geq 9} P_r^{-1} \approx 5 \times 10^{-9}$. This is a heuristic, not a proof: it assumes the cofactors behave like random integers, which is exactly what the arithmetic of exceptional valuations could violate (for instance $7^2 + 1 = 2 \cdot 5^2$ in the one known example).

7 Computational evidence

All computations used Python 3.13 with `gmpy2` and `sympy`, and PARI/GP 2.17.2 through `cypari2`.

(E1) *Direct search over y.* Both signs, $y \leq 2 \times 10^7$ for $k = 3, 4, 5, 6$, and $y \leq 10^6$ for $7 \leq k \leq 64$. Only $7^4 - 1$.

(E2) *All prime-complete numbers below 10^{30}.* Every prime-complete $N \leq 10^{30}$ (459,890,353 numbers) was enumerated by exponent vector, and $N \pm 1$ tested for being a perfect k-th power with $k \geq 3$. Only $N = 2400$. As a control, testing $N + 1$ for squares recovers all 28 terms of A141399. The program extends the A141399 generator of Chai Wah Wu [2].

(E3) *Exact classification for $k = 3$, $r \leq 8$* (Theorem 4.3), with an independent direct-search check.

(E4) *Cubic floor* computed exactly through $r = 32$ (§4).

Table 5: Combined coverage: every y in range has been tested for both $y^k - 1$ and $y^k + 1$.

k	range of y	source
3	$y \leq 10^{10}$; all y at orders $r \leq 8$	(E2), (E3)
4	$y \leq 3.16 \times 10^7$	(E2)
5, 6	$y \leq 2 \times 10^7$	(E1)
$7 \leq k \leq 64$	$y \leq 10^6$	(E1)
$k \geq 65$	$y^k \leq 10^{30}$	(E2)
even k	$y^{k/2} \leq 9.49 \times 10^{29}$	Corollary 2.3

A magnitude bound mostly covers small orders: by Lemma 3.1, at order r a solution needs $y \geq Q_k(r) \approx P_r^{\delta_k}$. That is why (E3), which is exhaustive per order regardless of the size of y, is the stronger form of evidence for $k = 3$.

8 A programme for a proof

Step A: even k. By Proposition 2.2 this is the subcase of A141399 in which the Pell abscissa $x_n(D) = 2m + 1$ is a perfect power. Any proof of A141399 settles it. Independently, perfect powers in Lucas and Pell sequences are a classical subject (for squares, the equation $X^4 - DY^2 = 1$ [verify: Ljunggren; Bennett–Walsh]), which could give a direct treatment branch by branch; uniformity over the branches D is the same obstacle as in A141399.

Step B: odd $k \geq 5$. Theorem 5.2 gives finiteness under *abc*. Two unconditional deliverables are realistic: (i) exact classification at small orders by the Thue algorithm of §4 (degree k forms, k^r classes); (ii) a version of Theorem 5.2 uniform in k, which needs Zsigmondy's theorem [16] (a primitive prime of $\Phi_k(y)$ is $\equiv 1 \pmod{k}$, so $k < p_r$) together with Lemma 3.2 for moduli growing with r.

Step C: $k = 3$, the core.

C1. *Extend the exact classification.* Push the Thue computation of $L_3(r)$ to $r \approx 12$, certifying unconditionally as far as class-group certification allows. The floor side is already cheap through $r = 32$. This gives an order-by-order certificate, as `LC_Solver` does for A141399.

C2. *Prove the cubic floor bound* (F_3). By Lemma 3.1, $y = \pm 1 + Q_3(r)\,t$, and the conditions on t live modulo $M_r = \prod_{q \leq p_r,\, q \equiv 1(3)} q \approx P_r^{1/2}$, in $3^{\pi(p_r;3,1)} = P_r^{o(1)}$ admissible classes. (F_3) asks only that none of these classes has a representative below P_r^c, far below the expected $P_r^{1/2-o(1)}$. It is a least-solution problem for an explicit CRT set, the cubic analogue of the floor problem for A118478, and is weaker than full equidistribution. With *abc*, it proves the case $k = 3$ (Theorem 5.4).

C3. *An unconditional attack on the critical stratum.* By Corollary 3.4, $\Phi_3(y)$ must be nearly powerful on the class 1 mod 3. Writing $\Phi_3(y) = cz^2$ with c squarefree turns this into the family of equations $(2y+1)^2 - 4cz^2 = -3$, indexed by squarefree c supported on $\{3\} \cup \{q \equiv 1\ (3)\}$. The condition $Q_3(r) \mid y-1$ is then a set of rank-of-apparition conditions on each family, which is the structure that the anchor theorem exploits for A141399. The same uniformity barrier is expected, but the extra congruence $y \equiv 1 \pmod{Q_3(r)}$, which A141399 lacks, is a new lever.

Step D: the general principle. The obstruction of §3 applies to every intersective polynomial with a non-linear irreducible factor: primes whose Frobenius has no fixed point on the roots of that factor are forced into the other factors. Formulating a general "residue-class tax" conjecture, and testing it on families such as $x(x^2+1)$ and $(x^2-2)(x^2-3)(x^2-6)$, is a natural next step.

Code and data. `pc_power_neighbors.py` (E2), `thue_k3.py` (E3) and `cubic_floor2.py` (E4) reproduce every number in this note.

References

[1] OEIS Foundation Inc., *A055932: numbers where all prime divisors are consecutive primes starting at 2*, https://oeis.org/A055932.

[2] OEIS Foundation Inc., *A141399*, https://oeis.org/A141399 (including the generator program of Chai Wah Wu, 5 April 2026, and the comments of K. Clements, 6 August 2026).

[3] M. A. Bennett, *Rational approximation to algebraic numbers of small height: the Diophantine equation* $|ax^n - by^n| = 1$, J. Reine Angew. Math. **535** (2001), 1–49.

[4] Yu. Bilu and G. Hanrot, *Solving Thue equations of high degree*, J. Number Theory **60** (1996), 373–392.

[5] N. G. de Bruijn, *On the number of positive integers $\leq x$ and free of prime factors $> y$, II*, Indag. Math. **28** (1966), 239–247. [verify: volume and pages]

[6] N. D. Elkies, *ABC implies Mordell*, Internat. Math. Res. Notices **1991**, no. 7, 99–109.

[7] A. Granville, *Smooth numbers: computational number theory and beyond*, in: Algorithmic Number Theory, MSRI Publ. **44**, Cambridge Univ. Press, 2008, 267–323.

[8] J. C. Lagarias and K. Soundararajan, *Smooth solutions to the abc equation: the xyz conjecture*, J. Théor. Nombres Bordeaux **23** (2011), 209–234.

[9] M. Langevin, *Cas d'égalité pour le théorème de Mason et applications de la conjecture (abc)*, C. R. Acad. Sci. Paris Sér. I **317** (1993), 441–444.

[10] D. H. Lehmer, *On a problem of Størmer*, Illinois J. Math. **8** (1964), 57–79.

[11] D. W. Masser, *Open problems*, Proc. Symp. Analytic Number Theory, Imperial College, London, 1985.

[12] L. J. Mordell, *Diophantine Equations*, Academic Press, 1969.

[13] C. Nelson, D. E. Penney and C. Pomerance, *714 and 715*, J. Recreational Math. **7** (1974), 87–89.

[14] J. Oesterlé, *Nouvelles approches du "théorème" de Fermat*, Sém. Bourbaki no. 694, Astérisque **161–162** (1988), 165–186.

[15] The PARI Group, *PARI/GP version 2.17.2*, Univ. Bordeaux, `https://pari.math.u-bordeaux.fr/`.

[16] K. Zsigmondy, *Zur Theorie der Potenzreste*, Monatsh. Math. Phys. **3** (1892), 265–284.